

PRODUCT REQUIREMENTS DOCUMENT

Cargo-Aware Drone Selection DSS for Small Ports

Group 6 — DSS University Project
Blekinge Institute of Technology
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1. Product Overview

1.1 Product Name

Cargo-Aware Drone Selection Decision Support System (DSS) for Small Ports

1.2 Short Description

This system is a web-based Decision Support System designed to help port operators choose the most suitable drone for a given mission. It filters out unsafe or infeasible drones using rule-based logic, ranks the remaining candidates using weighted decision criteria, and presents a clear recommendation with full explanations. The system also logs every decision for accountability and future reference.

1.3 Problem Statement

Small ports rely on drones for cargo inspection, perimeter surveillance, and operational monitoring. However, drone selection is still performed manually. Operators must simultaneously evaluate battery level, payload capacity, weather conditions, cargo zone distance, sensor availability, maintenance status, safety constraints, and budget — all under time pressure. This manual process leads to inconsistent decisions, potential safety risks, delayed missions, and inefficient use of drone resources.

1.4 Product Goal

Develop a practical, student-level DSS that:

- Automates drone feasibility checking through rule-based filtering.
- Ranks suitable drones using a transparent multi-criteria scoring method.
- Provides a clear recommendation with reasons visible to the operator.
- Supports mission scheduling suggestions and safety alerting.
- Maintains a full decision log for review and accountability.

1.5 Target Users

The primary users of this system are port operations staff at small to medium-sized ports who manage drone fleets for inspection and surveillance tasks. Secondary users include maintenance personnel and system administrators.

1.6 Student-Level Project Scope

This is a university DSS project. The system uses simulated or sample drone and environmental data. Real drone hardware control is out of scope. The focus is on decision support logic, user interface clarity, and proper data management. The technology stack is intentionally kept simple: Python, FastAPI, SQLite or PostgreSQL, and Streamlit.

2. Objectives

2.1 Main Objectives

1. Provide operators with a structured tool to select the most appropriate drone for each mission.
2. Eliminate unsafe or infeasible drone options automatically using pre-defined safety rules.
3. Rank remaining drones using configurable weighted scoring.
4. Show clear reasons for every recommendation and rejection.
5. Generate safety alerts when environmental or operational thresholds are exceeded.
6. Record every decision in a log for review and analysis.
7. Present all outputs through a clean, simple dashboard.

2.2 Problem the DSS Solves

Without the DSS, operators must manually cross-reference dozens of parameters to select a drone for each mission. This is slow, error-prone, and inconsistent across operators. The DSS standardises this process by applying the same rules and scoring criteria every time, reducing cognitive load and improving decision quality.

2.3 Expected Benefits

Benefit	Description
Faster Selection	Operators receive a ranked recommendation within seconds instead of manually comparing drones.
Safer Operations	Rule-based filtering prevents deployment of drones that fail safety or feasibility checks.
Better Planning	Mission scheduling suggestions help operators sequence missions efficiently.
Reduced Manual Workload	Automated filtering and scoring reduce the operator's cognitive load.
Improved Resource Usage	Scoring criteria include historical reliability and maintenance status, promoting better drone lifecycle management.
Accountability	Every decision is stored in a log with reasons, enabling audit and review.

3. User Personas

3.1 Port Operator

Attribute	Detail
Role	Primary daily user of the DSS
Goal	Quickly select the right drone and plan missions safely
Frustration	Too many parameters to evaluate manually under time pressure
System Use	Creates missions, reviews drone recommendations, confirms or overrides selections, checks alerts
Technical Level	Moderate — familiar with port operations software but not a developer

3.2 Port Security Supervisor

Attribute	Detail
Role	Reviews high-risk missions and provides override approval
Goal	Ensure all drone operations meet safety and regulatory standards
Frustration	Lack of visibility into operator decisions and safety evaluations
System Use	Views safety alerts, reviews decision logs, approves high-risk mission overrides
Technical Level	Moderate — reads reports and dashboards but does not configure the system

3.3 Maintenance Staff

Attribute	Detail
Role	Keeps drone data accurate and up to date
Goal	Ensure that drone status and maintenance records are current
Frustration	Incorrect drone status leading to unsafe deployments
System Use	Updates drone maintenance status, battery condition, and sensor availability in the drone inventory
Technical Level	Low to moderate — uses simple forms to update records

3.4 System Administrator

Attribute	Detail
Role	Manages system configuration and user accounts
Goal	Keep the system running correctly and adjust criteria weights when needed
Frustration	Lack of control over system settings from the UI
System Use	Manages user roles, adjusts decision criteria weights, maintains drone and cargo zone records
Technical Level	High — comfortable with backend settings and database management

4. Scope

4.1 In Scope — MVP

- Drone data management: add, edit, and view drone records.
- Mission creation: define mission type, urgency, cargo zone, distance, sensors needed, and flight height.
- Cargo zone input: manage zone location, obstacle density, safety risk, and cargo priority.
- Weather and environmental input: manual entry or mocked/simulated data.
- Rule-based filtering: automatically reject drones that fail safety or feasibility checks.
- Drone ranking: score remaining drones using weighted multi-criteria logic.
- Recommendation display: show the best drone with score and explanation.
- Rejected drone display: show which drones were rejected and why.
- Mission scheduling suggestion: suggest a time slot or sequence based on drone availability.
- Safety alert display: highlight conditions that exceed safe thresholds.
- Operator override: allow operator to override a recommendation with a mandatory reason.
- Decision log: store all decisions, scores, alerts, and overrides.
- Dashboard overview: show drone count, active missions, alerts, and recent decisions.
- Simple role selection or login (no enterprise authentication needed).

4.2 Out of Scope

- Real autonomous drone control or flight command dispatch.
- Real-time large-scale IoT or telemetry data streams from physical drones.
- Full computer vision or image processing pipeline.
- Reinforcement learning or LSTM-based models in the MVP.
- Complex cloud infrastructure, microservices, or Kubernetes deployment.
- Digital twin layer or 3D simulation.
- Real port hardware or SCADA integration.
- Enterprise-grade authentication with OAuth or SSO.
- Multi-port or multi-tenant deployment.

4.3 Future Enhancements

- Real weather API integration (e.g., OpenWeatherMap).
- Live telemetry feed with drone battery and location tracking.
- Computer vision module for cargo inspection from drone camera feeds.
- LSTM-based anomaly detection for flight stability monitoring.
- OR-Tools for optimal mission scheduling and resource allocation.
- Predictive maintenance based on flight hours and historical data.
- Role-based access control with secure authentication.

5. Functional Requirements

ID	Category	Requirement Description	Priority
FR-01	Access	The system shall allow users to select a role (Operator, Supervisor, Maintenance, Admin) from a simple login or role selection screen.	High
FR-02	Drone Management	The system shall allow authorised users to add new drone records with all capability fields.	High
FR-03	Drone Management	The system shall allow users to edit existing drone records, including maintenance status and battery level.	High
FR-04	Drone Management	The system shall display a list of all drones with key attributes and current status.	High
FR-05	Mission Management	The system shall allow operators to create a new mission by specifying type, urgency, cargo zone, distance, required flight height, and required sensors.	High
FR-06	Mission Management	The system shall allow users to view existing missions and their status.	Medium
FR-07	Mission Management	The system shall allow users to edit a mission before it is executed.	Medium
FR-08	Cargo Zone	The system shall allow users to add and view cargo zone records including location, obstacle density, safety risk, and cargo priority.	High
FR-09	Environmental Input	The system shall allow users to manually enter or load simulated weather and environmental conditions for a mission.	High
FR-10	Decision Engine	The system shall apply rule-based filtering to remove drones that are unsafe or infeasible for the given mission.	High
FR-11	Decision Engine	The system shall display each rejected drone with a clear explanation of the rejection reason.	High
FR-12	Decision Engine	The system shall score remaining drones using a weighted multi-criteria formula.	High
FR-13	Decision Engine	The system shall rank suitable drones by their total score in descending order.	High
FR-14	Recommendation	The system shall highlight the top-ranked drone as the recommended selection.	High
FR-15	Recommendation	The system shall show the recommendation score and the reasons contributing to it.	High
FR-16	Scheduling	The system shall suggest a mission time slot based on drone availability and mission urgency.	Medium
FR-17	Safety	The system shall generate safety alerts when weather, wind, battery, or risk thresholds are exceeded.	High

FR-1 8	Override	The system shall allow an operator to override the recommendation and select a different drone, with a mandatory reason field.	High
FR-1 9	Decision Log	The system shall store every decision, including selected drone, score, rejected drones, alerts, and any operator override.	High
FR-2 0	Decision Log	The system shall display a searchable history of past decisions and mission outcomes.	Medium
FR-2 1	Settings	The system shall allow the administrator to adjust the weight of each decision criterion.	Medium

6. Non-Functional Requirements

ID	Category	Requirement	Measure
NFR-01	Usability	The dashboard and recommendation screens shall be understandable without training.	A new user can complete a mission creation and review a recommendation within 5 minutes.
NFR-02	Usability	All rejection reasons and recommendation explanations shall be written in plain language.	No technical jargon in user-facing messages.
NFR-03	Performance	The rule-based filtering and scoring shall complete within 3 seconds for a fleet of up to 50 drones.	Measured in local testing environment.
NFR-04	Performance	Dashboard pages shall load within 5 seconds under normal conditions.	Measured in Streamlit local deployment.
NFR-05	Reliability	The system shall not crash during a standard mission creation and recommendation workflow.	Zero crashes in a 30-minute demonstration session.
NFR-06	Reliability	Decision logs shall persist across application restarts.	Data stored in database and verifiable after restart.
NFR-07	Maintainability	Decision rules shall be defined in a separate, editable configuration file or database table.	A rule can be updated without modifying core application code.
NFR-08	Maintainability	Code shall follow standard Python conventions and include basic comments.	Readable by another student without additional documentation.
NFR-09	Security	The system shall not expose raw database credentials in the UI or API responses.	No sensitive data visible in Streamlit or API output.
NFR-10	Explainability	Every recommendation shall be accompanied by a score breakdown showing contribution of each criterion.	Score breakdown visible on the recommendation page.
NFR-11	Data Accuracy	All drone and mission data entered by the user shall be validated before processing.	Form validation prevents empty or out-of-range values.
NFR-12	Scalability	The system shall support a drone fleet of up to 50 drones and up to 200 mission records for the student project.	Tested with sample data at this scale.

7. Decision Criteria

The DSS evaluates each drone against the mission requirements using the following criteria, organised into five groups. Each criterion is assigned a weight during scoring; weights can be adjusted by the administrator.

7.1 Drone Capability Criteria

Sub-total weight: 52%

#	Criterion	Description	Default Weight
C-0 1	Flight Radius	Maximum distance the drone can travel from its base on a single charge.	8%
C-0 2	Battery Endurance	Total flight time available on a fully charged battery.	11%
C-0 3	Payload Capacity	Maximum weight the drone can carry (relevant if cargo samples are transported).	6%
C-0 4	Camera Quality	Resolution and image quality of the onboard camera (HD, 4K, etc.).	5%
C-0 5	Night Vision Support	Whether the drone supports low-light or night vision operation.	4%
C-0 6	Thermal Sensor	Whether the drone carries a thermal imaging sensor.	4%
C-0 7	Flight Height Capability	Maximum altitude the drone can safely reach.	4%
C-0 8	Wind Resistance Level	Maximum wind speed the drone can operate in.	8%
C-0 9	Charging Time	Time required to fully recharge the drone battery.	2%

7.2 Mission Requirement Criteria

Sub-total weight: 20%

#	Criterion	Description	Default Weight
C-1 0	Mission Urgency	How time-critical the mission is (Low, Medium, High, Emergency).	8%
C-1 1	Cargo Zone Distance	Distance from the drone base to the target cargo zone.	6%

C-1 2	Historical Reliability	Past performance score based on mission success rate and incident history.	6%
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7.3 Environmental Condition Criteria

Sub-total weight: 15%

#	Criterion	Description	Default Weight
C-1 3	Weather Condition	Current weather state: clear, cloudy, rain, storm (affects overall feasibility).	5%
C-1 4	Wind Speed	Current wind speed at the operating location (distinct from the drone's wind resistance rating).	7%
C-1 5	Obstacle Density	Density of physical obstacles (cranes, containers, structures) in the cargo zone.	3%

7.4 Safety Constraint Criteria

Sub-total weight: 10% (Maintenance Status is a hard elimination rule, not a scored weight)

#	Criterion	Description	Default Weight
C-1 6	Safety Risk Level	Overall mission risk classification (Low, Medium, High, Critical).	6%
C-1 7	Regulatory Compliance	Whether the drone holds the required operating certifications for the mission.	4%
C-1 8	Maintenance Status	Current maintenance state of the drone (Good, Scheduled, Poor, Grounded).	Hard Rule — not scored

7.5 Operational Constraint Criteria

Sub-total weight: 3% (Available Operators is a hard elimination rule, not a scored weight)

#	Criterion	Description	Default Weight
C-1 9	Number of Daily Missions	How many missions the drone has already completed today (affects battery fatigue).	2%
C-2 0	Available Operators	Whether a qualified operator is available to supervise the mission.	Hard Rule — not scored

C-2 1	Budget Limitation	Operating cost of the drone per mission relative to available budget.	1%
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Total scored weight across all non-hard-rule criteria: 52% + 20% + 15% + 10% + 3% = 100%

8. Decision Logic

8.1 Process Overview

The DSS follows a two-phase approach: first it eliminates drones that are unsafe or infeasible (hard filtering), then it scores and ranks the remaining drones (soft scoring). This ensures that only safe and feasible drones are evaluated against the mission criteria.

8.2 Step-by-Step Decision Process

Step	Action	Description
1	Collect Input Data	Gather mission details, cargo zone data, selected environmental conditions, and the current drone fleet status from the database.
2	Apply Rule-Based Filtering	Run each drone through the hard-rule filter. Any drone that fails one or more rules is immediately rejected.
3	Record Rejection Reasons	For each rejected drone, record the specific rule(s) it failed for display to the operator.
4	Score Remaining Drones	Apply the weighted multi-criteria formula to all drones that passed filtering.
5	Rank Drones	Sort drones by total score in descending order.
6	Recommend Best Drone	Present the highest-scoring drone as the recommendation.
7	Display Explanation	Show the score breakdown, rejection list with reasons, and any active safety alerts.
8	Save Decision Log	Store the full decision (inputs, scores, recommendation, rejections, alerts, and any override) in the database.

8.3 Hard Filter Rules (Automatic Rejection)

Rule ID	Condition	Rejection Message
R-01	Drone battery endurance < required mission flight time × 1.2 (20% safety margin)	Insufficient battery endurance for this mission distance.
R-02	Current wind speed > drone maximum wind resistance rating	Wind speed exceeds drone wind resistance limit.
R-03	Mission requires night vision AND drone does not support night vision	Required night vision sensor not available.

R-04	Mission requires thermal sensor AND drone does not have thermal sensor	Required thermal sensor not available.
R-05	Drone maintenance status = 'Poor' or 'Grounded'	Drone is not cleared for operation due to maintenance status.
R-06	Cargo zone distance > drone maximum flight radius	Cargo zone is beyond the drone's operational range.
R-07	Mission safety risk = 'Critical' AND no supervisor approval on record	Mission safety risk is Critical. Supervisor approval required before deployment.
R-08	Drone regulatory compliance = False for the mission type	Drone does not meet regulatory requirements for this mission type.
R-09	No qualified operator is available	No operator is currently available to supervise this drone.
R-10	Weather condition = Storm or Heavy Rain	Current weather conditions are unsafe for drone operations.

8.4 Scoring Formula

Each drone that passes the hard filter receives a total score calculated as a weighted sum of normalised criterion scores:

$$\text{Total Score} = \sum (\text{Criterion Weight} \times \text{Normalised Criterion Score})$$

Each criterion score is normalised to a scale of 0 to 1. Weights are expressed as decimals summing to 1.0. The result is a final score between 0 and 1 for each drone, which determines the ranking.

Example: A drone with battery endurance score of 0.9, wind resistance score of 0.8, and all other scores averaged at 0.7 will receive a total score that reflects the configured weights. The administrator can adjust weights from the Settings page.

9. Data Requirements

9.1 Drone

Field	Type	Description
drone_id	String (PK)	Unique identifier for the drone
drone_name	String	Display name or model name
flight_radius_km	Float	Maximum operational range in kilometres
battery_endurance_min	Integer	Maximum flight time in minutes on full charge
payload_capacity_kg	Float	Maximum payload in kilograms
camera_quality	Enum	Camera resolution: SD, HD, 4K
night_vision	Boolean	Supports night vision operation
thermal_sensor	Boolean	Equipped with thermal imaging sensor
max_flight_height_m	Integer	Maximum operational altitude in metres
wind_resistance_kmh	Float	Maximum tolerable wind speed in km/h
charging_time_min	Integer	Full recharge time in minutes
maintenance_status	Enum	Good, Scheduled, Poor, Grounded
regulatory_compliant	Boolean	Holds required operating certification
reliability_score	Float (0–1)	Historical mission success rate
daily_missions_done	Integer	Number of missions completed today
last_updated	Timestamp	Last record update timestamp

9.2 Mission

Field	Type	Description
mission_id	String (PK)	Unique mission identifier
mission_type	Enum	Inspection, Surveillance, Monitoring, Emergency
urgency_level	Enum	Low, Medium, High, Emergency
cargo_zone_id	String (FK)	Reference to the target cargo zone
required_distance_km	Float	Expected flight distance in kilometres
required_height_m	Integer	Required operational altitude in metres
requires_night_vision	Boolean	Mission needs night vision capability
requires_thermal	Boolean	Mission needs thermal sensor
operator_id	String (FK)	Assigned operator (if confirmed)
assigned_drone_id	String (FK)	Selected drone after recommendation

scheduled_time	Timestamp	Suggested or confirmed mission start time
status	Enum	Draft, Scheduled, Active, Completed, Cancelled
created_at	Timestamp	Mission creation timestamp

9.3 Cargo Zone

Field	Type	Description
zone_id	String (PK)	Unique cargo zone identifier
zone_name	String	Display name of the cargo zone
location_description	String	Text description of zone location in the port
distance_from_base_k m	Float	Distance from drone base to zone
obstacle_density	Enum	Low, Medium, High
safety_risk_level	Enum	Low, Medium, High, Critical
cargo_priority	Enum	Standard, Priority, Hazardous
inspection_requiremen ts	Text	Notes on specific inspection needs

9.4 Environmental Data

Field	Type	Description
env_id	String (PK)	Unique environmental record identifier
mission_id	String (FK)	Associated mission
weather_condition	Enum	Clear, Cloudy, Rain, Heavy Rain, Storm
wind_speed_kmh	Float	Wind speed at operating location in km/h
visibility_m	Integer	Horizontal visibility in metres
temperature_c	Float	Ambient temperature in Celsius
recorded_at	Timestamp	Time of environmental reading or manual entry
data_source	Enum	Manual, Simulated, API

9.5 Telemetry Data (Simulated)

Field	Type	Description
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telemetry_id	String (PK)	Unique telemetry record identifier
drone_id	String (FK)	Associated drone
mission_id	String (FK)	Associated mission
battery_pct	Integer	Current battery percentage (0–100)
location_lat	Float	Current latitude coordinate
location_lon	Float	Current longitude coordinate
altitude_m	Float	Current altitude in metres
speed_kmh	Float	Current flight speed in km/h
sensor_status	String	Status of onboard sensors
recorded_at	Timestamp	Timestamp of the telemetry reading

9.6 Decision Log

Field	Type	Description
log_id	String (PK)	Unique decision log identifier
mission_id	String (FK)	Related mission
recommended_drone_id	String (FK)	Drone recommended by the DSS
recommendation_score	Float	Total score of the recommended drone
score_breakdown	JSON	Criterion-level score breakdown
rejected_drones	JSON	List of rejected drones and their rejection reasons
safety_alerts	JSON	List of safety alerts generated
operator_override	Boolean	Whether the operator overrode the recommendation
override_drone_id	String (FK)	Drone selected if operator overrode
override_reason	Text	Reason provided by the operator for the override
decided_by	String (FK)	Operator or role who confirmed the decision
decision_timestamp	Timestamp	When the decision was made

9.7 User / Operator

Field	Type	Description
user_id	String (PK)	Unique user identifier
username	String	Login name

full_name	String	User's full name
role	Enum	Operator, Supervisor, Maintenance, Admin
is_available	Boolean	Whether the operator is currently on duty
created_at	Timestamp	Account creation timestamp

10. User Interface Requirements

10.1 General UI Principles

- Built using Streamlit for the MVP; a lightweight, Python-native framework.
- Left sidebar navigation with clearly labelled page links.
- Minimal colour palette: white background, navy/dark blue accents, green for success, amber for warning, red for rejection or danger.
- Key figures displayed in metric cards at the top of each page.
- Tables used for drone comparison with sortable columns.
- Status badges used consistently: Suitable (green), Warning (amber), Rejected (red).
- Recommendation displayed as a prominent highlighted card with score and reasoning.
- Rejected drones shown in a separate table below the recommendation.
- No crowded screens; each page has one clear primary purpose.
- Plain language throughout; no technical jargon in user-facing labels.

10.2 Page Inventory

Page	Primary Purpose	Key Components
Dashboard Overview	At-a-glance operational status	Metric cards, weather status, recent alerts, quick action buttons
Drone Inventory	View and manage drone fleet	Sortable drone table, status badges, add/edit forms
Create Mission	Define a new mission	Input form for all mission parameters, submit button
Drone Recommendation	Review DSS recommendation	Best drone card, score breakdown, ranked table, rejected table, confirm/override action
Mission Schedule	View upcoming and active missions	Timeline or list view, status colours, assigned drone info
Alerts & Safety Warnings	Review active and past safety alerts	Alert list with severity, timestamp, and related mission
Decision Logs	Audit past decisions	Searchable log table, detail expand view, override indicator
Settings / Weights	Configure decision criteria weights	Sliders or number inputs for each criterion weight, save button

11. Clean UI Layout Suggestion

11.1 Dashboard Overview

Top row — metric cards:

- Total Drones | Available Drones | Active Missions | High-Risk Missions

Second row:

- Current Weather Status card (condition, wind speed, visibility).
- Recent Alerts panel (last 5 alerts with severity badge and timestamp).

Bottom row:

- Recent Decisions table (last 5 decisions: mission, drone, score, operator).
- Quick action buttons: New Mission, Update Drone Status.

11.2 Drone Inventory Page

- Search and filter bar at the top (filter by status, maintenance, sensor support).
- Drone table with columns: Drone ID, Name, Range, Battery, Payload, Wind Resistance, Maintenance Status, Reliability Score, Action buttons (Edit, View).
- Status badge in Maintenance Status column: Good (green), Scheduled (amber), Poor (red), Grounded (dark red).
- Add New Drone button that opens a form (Streamlit form with all drone fields).

11.3 Create Mission Page

- Mission Type (dropdown: Inspection, Surveillance, Monitoring, Emergency).
- Urgency Level (dropdown: Low, Medium, High, Emergency).
- Cargo Zone (dropdown linked to cargo zone records).
- Required Distance km (number input, auto-filled if cargo zone is selected).
- Required Flight Height m (number input).
- Requires Night Vision (checkbox).
- Requires Thermal Sensor (checkbox).
- Weather Condition (dropdown: manual or load simulated).
- Wind Speed km/h (number input).
- Submit button → triggers the decision engine and navigates to Recommendation page.

11.4 Drone Recommendation Page

- Top: green highlighted card showing the best drone — Drone Name, Total Score (e.g. 0.84), key reason summary.
- Score Breakdown section: bar chart or table showing each criterion's contribution.
- Ranked Drone Table: all suitable drones with individual scores, sortable.
- Rejected Drone Table: drone name, rejection rule(s), shown in red-tinted rows.
- Active Safety Alerts section: amber/red alert cards if any thresholds exceeded.
- Action buttons: Confirm Recommendation | Override (opens a modal with mandatory reason text field).

11.5 Mission Schedule Page

- List or simple timeline view of upcoming and active missions.
- Each row: Mission ID, Type, Urgency badge, Cargo Zone, Assigned Drone, Scheduled Time, Status badge.
- Colour-coded status: Draft (grey), Scheduled (blue), Active (green), Completed (teal), Cancelled (red).

11.6 Alerts & Safety Warnings Page

- Filter controls: by severity, date range, or mission.
- Alert list: each alert shows severity badge (Info, Warning, Critical), description, related mission, and timestamp.
- Click to expand shows full alert context including which rule was triggered.

11.7 Decision Logs Page

- Search by mission ID, date, operator, or drone.
- Log table: Mission ID, Recommended Drone, Score, Override (Yes/No), Operator, Timestamp.
- Click to expand row: shows score breakdown, rejected drones list, safety alerts, override reason if applicable.

11.8 Settings / Criteria Weights Page

- List of all criteria with current weight (as percentage).
- Slider or number input per criterion; total must sum to 100%.
- Live total indicator showing current sum.
- Save Weights button with confirmation message.

12. User Stories

ID	User Story	Acceptance Summary
US-01	As a Port Operator, I want to create a new mission by entering mission type, urgency, cargo zone, and environmental conditions, so that the system can evaluate suitable drones for me.	Mission creation form accepts all required fields and triggers the decision engine on submission.
US-02	As a Port Operator, I want to see a ranked list of suitable drones with scores, so that I can understand why a drone is recommended.	Ranked table with scores and score breakdown is displayed on the recommendation page.
US-03	As a Port Operator, I want to see which drones were rejected and the exact reasons, so that I understand the limitations of my fleet.	Rejected drones table with rule-based rejection messages is visible below the recommendation.
US-04	As a Port Operator, I want to override the DSS recommendation and select a different drone, so that I retain final control over the decision.	Override option is available; operator must enter a reason before confirming the override.
US-05	As a Port Security Supervisor, I want to review safety alerts for active and recent missions, so that I can ensure all operations meet safety standards.	Alerts page shows all alerts with severity, description, and related mission.
US-06	As a Port Security Supervisor, I want to see the decision log for any mission, so that I can audit what drone was selected and why.	Decision log shows recommendation, score breakdown, rejected options, and override reason if applicable.
US-07	As Maintenance Staff, I want to update the maintenance status and battery condition of a drone, so that the DSS uses accurate data when making recommendations.	Drone edit form allows updating maintenance status; changes are reflected immediately in the next DSS evaluation.
US-08	As a System Administrator, I want to adjust the weight of each decision criterion from the Settings page, so that the scoring reflects current operational priorities.	Settings page allows weight adjustment; weights sum check is enforced; save is confirmed with a message.
US-09	As a Port Operator, I want to view my mission history and the decisions made, so that I can learn from past operations and improve future planning.	Decision logs page shows searchable history of past decisions with expandable detail.
US-10	As a Port Operator, I want to see a mission scheduling suggestion after a drone is recommended, so that I can plan the mission at an appropriate time.	Scheduling page or recommendation page shows a suggested time slot based on drone availability and urgency.
US-11	As a Port Security Supervisor, I want Critical-risk missions to require approval before a drone is confirmed, so that high-risk operations are never deployed without oversight.	If safety risk is Critical, the system blocks confirmation and prompts supervisor approval.

13. Acceptance Criteria

Feature	Acceptance Criteria
Drone Management	1. A new drone can be added with all required fields. 2. An existing drone can be edited. 3. The drone list displays all drones with current status. 4. Invalid or missing fields are rejected with a clear error message.
Mission Creation	1. All required mission fields are present and validated. 2. Submitting the form triggers the decision engine. 3. Missing mandatory fields prevent form submission.
Rule-Based Filtering	1. A drone with battery endurance below the required threshold is rejected. 2. A drone with wind resistance below current wind speed is rejected. 3. A drone with maintenance status of 'Grounded' is rejected. 4. Each rejected drone displays the exact rule it failed.
Drone Scoring and Ranking	1. All drones that pass filtering receive a numeric score. 2. Drones are listed in descending score order. 3. Score breakdown per criterion is visible to the operator.
Recommendation Display	1. The top-ranked drone is visually highlighted as the recommendation. 2. The recommendation card shows drone name, score, and key reasons. 3. Rejected drones are displayed separately with reasons.
Safety Alerts	1. An alert is generated when wind speed exceeds 80% of a drone's wind resistance. 2. An alert is generated when weather condition is Heavy Rain or Storm. 3. Alerts are visible on both the recommendation page and the alerts page.
Operator Override	1. The operator can select a different drone from the ranked list. 2. The override reason field is mandatory; submission is blocked without it. 3. The override is recorded in the decision log.
Decision Log	1. Every confirmed decision is stored in the database. 2. The log shows mission, recommended drone, score, operator, and timestamp. 3. Expanding a log entry shows score breakdown and any override details.
Settings / Weights	1. All criterion weights are displayed and editable. 2. The system warns if weights do not sum to 100%. 3. Saved weights are used in subsequent scoring calculations.
Dashboard	1. The dashboard displays total drones, available drones, active missions, and high-risk missions. 2. Recent alerts are visible. 3. The page loads without errors.

14. Technical Requirements

14.1 Recommended Tech Stack

Layer	Technology	Purpose	Status
Core Logic	Python 3.10+	DSS filtering, scoring, and data processing.	Required
Backend API	FastAPI	REST API connecting the UI to the decision engine and database.	Required
Database	SQLite (dev) / PostgreSQL (optional)	Persistent storage of drone, mission, and log data.	Required
Data Processing	Pandas, NumPy	Data manipulation, normalisation, and scoring calculations.	Required
Frontend / UI	Streamlit	Operator dashboard and all user-facing pages.	Required
API Testing	Postman	Manual testing of all FastAPI endpoints.	Required
Version Control	GitHub	Code collaboration and version tracking.	Required
Containerisation	Docker	Optional: consistent local development environment.	Optional
ML Models	PyTorch / scikit-learn	Optional future enhancement only; not in MVP.	Future
Scheduling	OR-Tools	Optional future enhancement for optimised scheduling.	Future
Computer Vision	OpenCV / YOLO	Optional future enhancement for cargo inspection.	Future

14.2 API Design Principles

- All endpoints follow REST conventions (GET for retrieval, POST for creation, PUT for updates).
- Request and response bodies use JSON format.
- Validation errors return HTTP 422 with descriptive messages.
- Decision engine is called via a POST endpoint (e.g., POST /decisions/evaluate) that accepts mission and environmental data.

15. System Architecture

15.1 Architecture Overview

The system is organised into five logical layers. Each layer has a clearly defined responsibility. There are no microservices; the entire backend runs as a single FastAPI application.

Layer	Components	Responsibility
Input / Data Collection Layer	Streamlit forms, manual weather entry, simulated telemetry loader	Capture mission parameters, drone updates, environmental conditions, and cargo zone details from the user or simulated sources.
Database Layer	SQLite or PostgreSQL via SQLAlchemy ORM	Persist all drone, mission, cargo zone, environmental, telemetry, decision log, and user data.
Decision Engine Layer	Python rule engine, scoring module, ranking module	Apply hard filter rules, calculate weighted scores, rank drones, generate safety alerts, and produce the recommendation.
Dashboard / UI Layer	Streamlit multi-page application	Display all information to users: dashboard, drone inventory, mission creation, recommendation, schedule, alerts, logs, and settings.
Decision Log / Report Layer	Decision log API endpoints, log database table	Store every decision and expose it for audit, review, and export.

15.2 Data Flow Summary

- Operator enters mission details in Streamlit → sent to FastAPI via HTTP POST.
- FastAPI retrieves current drone fleet, cargo zone, and environmental data from the database.
- Decision Engine applies hard filter rules → rejected drones recorded with reasons.
- Decision Engine scores and ranks remaining drones using weighted criteria.
- Result (recommendation, ranked list, rejections, alerts) returned to Streamlit.
- Operator reviews result, confirms or overrides → confirmation sent back to FastAPI.
- FastAPI writes the complete decision log entry to the database.

16. Risks and Mitigation

Risk	Likelihood	Impact	Mitigation Strategy
Lack of real drone or port data	High	Medium	Use a generated sample dataset of 20–30 realistic drone records and 10–15 cargo zones. Document clearly that data is simulated.
Weather API integration difficulty	Medium	Low	Use manual input or a simple mock JSON file for weather data in MVP. Real API integration is a future enhancement.
Decision criteria too complex to implement fully	Medium	High	Start with the 10 most impactful criteria in MVP. Remaining criteria can be added incrementally.
Time limitation of student project	High	High	Prioritise core filtering, scoring, and recommendation display. Dashboard and logs are secondary. Use agile sprints with weekly review.
Over-engineering the system	Medium	High	Follow the PRD strictly. Avoid adding features not listed in the MVP scope. Regularly review scope alignment.
UI becoming too crowded	Medium	Medium	Each page must have one primary purpose. Limit the number of visible elements. Use expandable sections for secondary details.
Team coordination issues	Medium	Medium	Use GitHub for version control. Assign clear ownership per module. Hold short weekly sync meetings.
Database schema changes mid-project	Low	Medium	Define and agree on the full schema before implementation begins. Use migration scripts if changes are needed.

17. Assumptions and Constraints

17.1 Assumptions

- Sample or simulated drone and cargo zone data will be used in place of real port data.
- Real drone hardware control is not implemented; the system only supports decision-making.
- Weather data can be entered manually or loaded from a simulated mock source; a real weather API is a future enhancement.
- The system is designed for decision support only; the operator always retains final control and responsibility.
- All users are assumed to be trusted internal port staff; advanced authentication is not required for the MVP.
- The system will be run locally or on a single server; multi-user concurrent access is not a primary requirement for the MVP.
- Drone and mission data accuracy depends on users keeping records up to date.

17.2 Constraints

- Technology stack is limited to Python, FastAPI, SQLite or PostgreSQL, Pandas, NumPy, and Streamlit.
- No real autonomous drone control or flight command dispatch is implemented.
- The system must be completable within the timeframe of a single university semester.
- No enterprise authentication, SSO, or role-based access control in the MVP.
- No real-time IoT data ingestion from physical drone hardware in the MVP.

18. MVP Success Criteria

The project is considered a successful student MVP when all of the following criteria are demonstrably met:

#	Success Criterion	Verification Method
S-01	A user can create a mission by entering all required parameters.	Live demonstration of mission creation form.
S-02	The system applies rule-based filtering and rejects unsafe drones with clear reasons.	Show a scenario where at least two drones are rejected for different rules.
S-03	The system scores and ranks the remaining drones using weighted criteria.	Ranked drone table is visible with scores.
S-04	The system recommends one drone with a score and explanation.	Best drone card with score breakdown is visible on recommendation page.
S-05	The operator can override the recommendation and enter a reason.	Override flow completes and reason is stored.
S-06	Safety alerts are generated and displayed when thresholds are exceeded.	Set wind speed above a drone's limit; alert appears.
S-07	Every decision is stored in the decision log and can be reviewed.	Decision log page shows the completed mission with all details.
S-08	The dashboard displays current fleet status and recent alerts.	Dashboard loads without errors and shows correct counts.
S-09	The administrator can adjust criterion weights and the scoring updates accordingly.	Change weights, re-run same mission, observe different ranking.

19. Future Enhancements

The following features are explicitly out of scope for the MVP but are identified as natural extensions of the system for future development:

Enhancement	Description	Technology
Real Weather API Integration	Replace manual weather input with live data from OpenWeatherMap or a similar service.	OpenWeatherMap API / Requests
Live Telemetry Feed	Ingest real-time battery, location, and sensor status from physical or simulated drone hardware.	MQTT, WebSocket, or MAVLink
Computer Vision for Cargo Inspection	Analyse drone camera feeds to detect cargo anomalies or hazardous goods automatically.	OpenCV, YOLO v8
LSTM-Based Anomaly Detection	Detect abnormal flight patterns or battery behaviour using a time-series model trained on telemetry data.	PyTorch, LSTM
OR-Tools Mission Scheduling	Optimise the sequencing and timing of multiple concurrent missions to minimise downtime and battery waste.	Google OR-Tools
Predictive Maintenance	Forecast when a drone is likely to require maintenance based on flight hours and historical incident data.	scikit-learn regression
Role-Based Authentication	Implement secure login with role-based access control so different users see only the features relevant to their role.	FastAPI OAuth2, JWT
Multi-Port Support	Extend the system to manage drone fleets across multiple port locations from a single instance.	Multi-tenant database design
Mobile-Responsive UI	Adapt the dashboard for use on tablets and smartphones used in field operations.	Streamlit or a React frontend

